Network Security
CS3403

Lab 2
Static NAT using CISCO Packet Tracer

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Lab 2: Focus

- Network Address Translation – Introduction
- Configuring the Hosts and Router Interfaces
- **Exp 0**: Configuring Static NAT
- **Exp 1**: Accessing the Web Server0 from PC1

File shared:
Lab2_NAT_Static_shared.pkt

Network Address Translation Introduction (NAT)

Using:
Lab2_NAT_Static_shared.pkt

Network Address Translation (NAT)

- Network Address Translation (NAT) is a technique used in networking to modify IP addresses in packet headers while they are sent out of a private network to public network (Internet).
  - The **Private source IP addresses** of the hosts are replaced with **Public IP addresses**
- It allows multiple devices on a private network to share a single or a limited set of public IP addresses to access the Internet.
  - Note: Using the Limited Public IP addresses allotted by ISP to an organization
  - Since IPv4 addresses are limited, NAT allows multiple devices to share public IP addresses.
- This also provides security by not exposing the private IP addresses of the hosts which are visible only within the organization

Static NAT: Explained

- Static NAT (Network Address Translation) refers to a network configuration where a specific private IP address on a local network is permanently mapped to a single public IP address
- It creates a one-to-one translation that allows devices on the private network to be directly accessible from the Internet using a different Public IP address.
- Commonly used for publicly accessible services like web servers, email servers, or remote access applications where a consistent external IP is needed.
- **Static NAT** is set up in a router or firewall, which is configured with a rule that specifies an internal IP address and the corresponding public IP address to be used for translation.

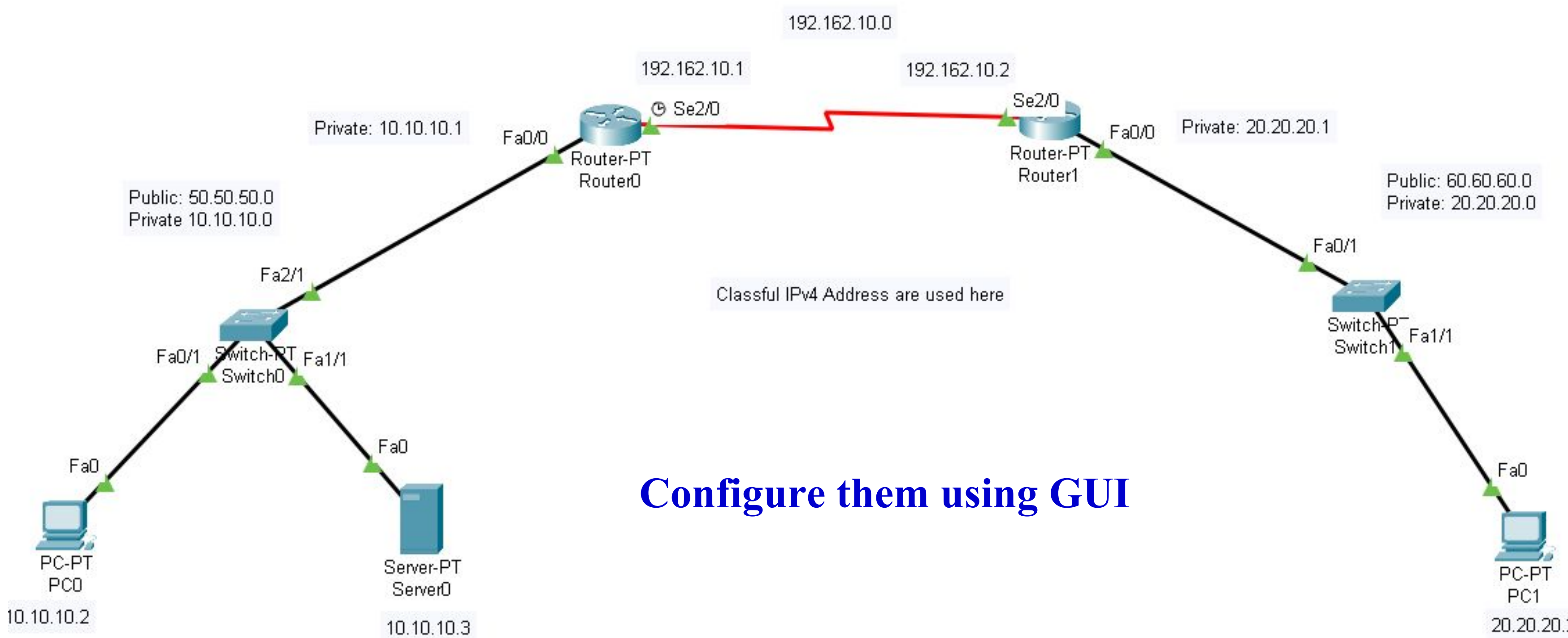
Network Address Translation Configuration (NAT)

Exp1: Static NAT Configuration using CLI

(File name: Lab5_Exp1_Static_NAT.pkt)

Using: Lab5_Exp1_Static_NAT_shared.pkt

Static NAT Demo: Exp1



Configure them using GUI

- **Step1**: Configure **PC0** and **Server0** with private IP addresses **10.10.10.2** and **10.10.10.3** and set its **default GW** as **10.10.10.1** (Set **Route0 Fa0/0** set **to 10.10.10.1**)
- **Step2**: Similarly, **PC1** with **20.20.20.2** and its **Default GW** as **20.20.20.1** (Set **Router1 Fa0/0** set to **20.20.20.1**
- **Step3:** Configure **Serial2/0** interfaces of the **Router0** and **Router1** as **192.162.10.1** and **192.162.10.2** and **enable** them as well**.**

Configure using *CLI*

Note: The **CLI commands** to be given are shown in ***blue bold and italicized***

Summary of Modes:

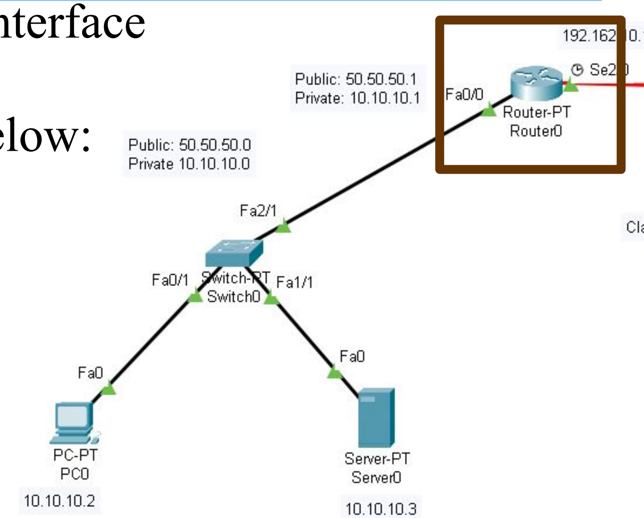
User mode: Router> → **View basic information**
Privilege mode: Router# *enable* → **Access config and diagnostics**
Global config mode: Switch (config)# *config terminal* → **Config device-wide settings**

Exp1: Static NAT Configuration using CLI

Configure Static NAT on Router0

- Let us configure the static NAT mapping using the **CLI** interface of **Router0**
- Need to **map** the **internal Private addresses** as shown below:
  - **10.10.10.2 □ 50.50.50.2**
  - **10.10.10.3 □ 50.50.50.3**
- **Step4: Router0 □ CLI □ Give** the **below commands**
  - **Router>** ***enable***
  - **Router#** ***show ip nat translations***
  - **Router#**

Note: No output



Inside global: Inside device's global IP addr
Inside local: Inside device's local IP addr

- **Step5:**
  - **Router#** ***config terminal***
  - **Router (config)#** ***ip nat inside source static 10.10.10.2 50.50.50.2***
  - **Router (config)#** ***ip nat inside source static 10.10.10.3 50.50.50.3***
  - **Router (config)#** ***exit***
- **Step6:**
  - **Router#** ***show ip nat translations***

Step6

```
Router#
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  50.50.50.2        10.10.10.2         ---                ---
---  50.50.50.3        10.10.10.3         ---                ---
```

Step4

```
Router>
Router>enable
Router#show ip nat translations
Router#
```

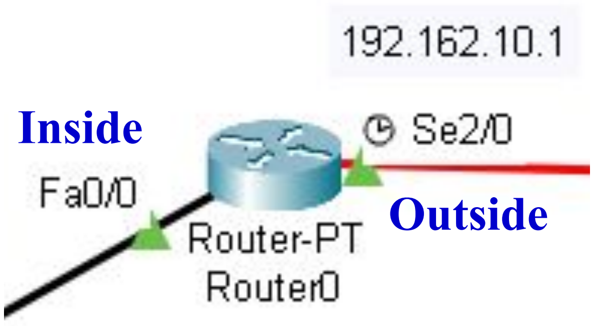
Step5

```
Router#
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat inside source static 10.10.10.2 50.50.50.2
Router(config)#ip nat inside source static 10.10.10.3 50.50.50.3
Router(config)#exit
Router#
```

Configure Router0 Interfaces

As Inside or Outside

- Need to configure which **interface** of the router is connected to the **inside** of the network (**private**) and the interface which is connected to the **outside** network (**Internet**):
  - **Step7:**
  - **Router#** ***config t***
  - **Router (config)#** ***interface fastEthernet 0/0***
  - **Router (config-if)#** ***ip nat inside***
  - **Router (config-if)#** ***exit***
  - **Router (config)#**
  - **Step8:**
  - **Router (config)#** ***interface serial 2/0***
  - **Router (config-if)#** ***ip nat outside***
  - **Router (config-if)#** ***exit***
  - **Step9:**
  - **Router (config)#*exit***
  - **Router#** ***show ip nat statistics***



Step9

```
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial2/0
Inside Interfaces: FastEthernet0/0
Hits: 0  Misses: 0
Expired translations: 0
Dynamic mappings:
Router#
```

Step7

```
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#
```

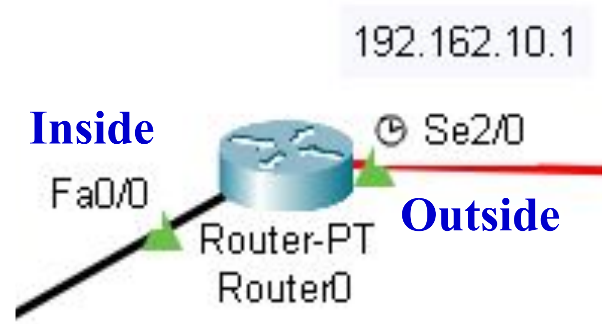
Step8

```
Router(config)#
Router(config)#interface serial 2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
```

Configure Router0 and Router1 Routing Tables

Configure Routing Tables using CLI

- Configure the **Routing table** on **Router0** using **CLI**
  - **Step10:**
  - **Router#** ***config t***
  - **Router (config)#** ***ip route 60.0.0.0 255.0.0.0 192.162.10.2***
  - **Router (config-if)#** ***exit***
  - **Router (config)#**



- Configure the **Routing table** on **Router1** using **CLI**
  - **Step11:**
  - **Router#** ***config t***
  - **Router (config)#** ***ip route 50.0.0.0 255.0.0.0 192.162.10.1***
  - **Router (config-if)#** ***exit***

Steps 12 & 13: ***show ip route*** on **Router0** and **Router1**

Step10

```
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#ip route 60.0.0.0  255.0.0.0 192.162.10.2
Router(config)#
```

Step11

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#ip route 50.0.0.0 255.0.0.0 192.162.10.1
Router(config)#
```

Step12

```
Router#
Router#show ip route
Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
S    60.0.0.0/8 [1/0] via 192.162.10.2
C    192.162.10.0/24 is directly connected, Serial2/0
```

Step13

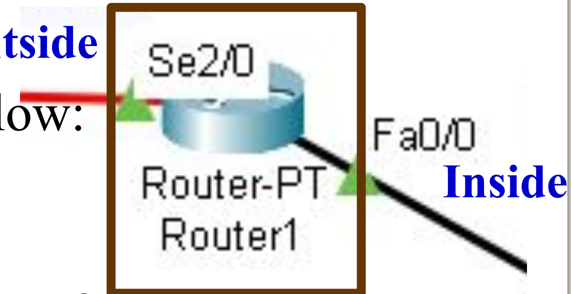
```
Router#
Router#show ip route
Gateway of last resort is not set

C    20.0.0.0/8 is directly connected, FastEthernet0/0
S    50.0.0.0/8 [1/0] via 192.162.10.1
C    192.162.10.0/24 is directly connected, Serial2/0
```

Exp1: Static NAT Configuration using CLI

Configure Static NAT on Router1

- Similarly configure the static NAT mapping using the **CLI** interface of **Router1**
- Need to **map** the **internal Private address** as shown below:
  - **20.20.20.2 □ 60.60.60.2**
- Configure the interfaces **Fa0/0** and **Se2/0** as inside and **outside** respectively, similar to what was done with **Router0**



- These **ping** commands should **succeed** from **PC0**
- ***ping 10.10.10.1***
- ***ping 10.10.10.2***
- ***ping 10.10.10.3***
- ***ping 192.162.10.1***
- ***ping 192.162.10.2***
- ***ping 60.60.60.2***

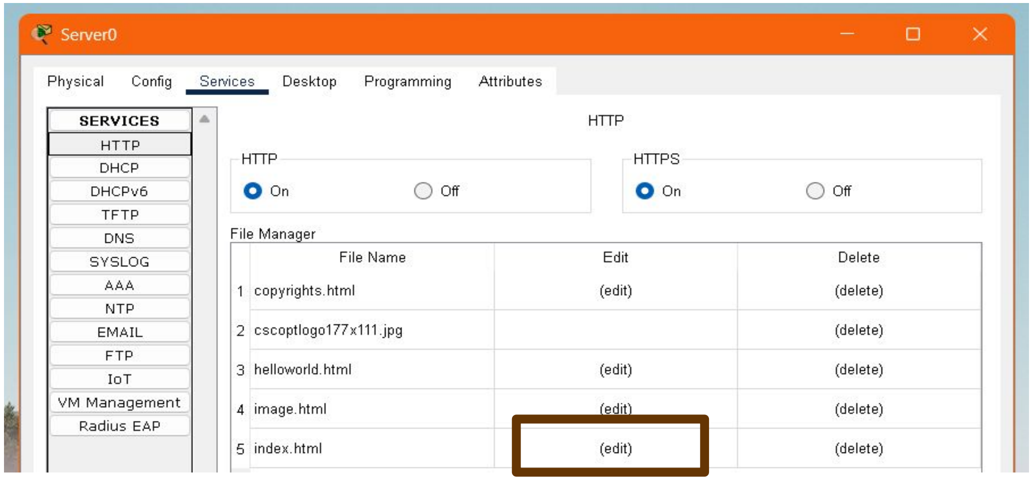
- These **ping** commands will **fail** from **PC0**
- ***ping 50.50.50.2***
- ***ping 50.50.50.3***
- ***ping 20.20.20.1***
- ***ping 20.20.20.2***
- These **ping** commands will **fail** from **PC1**
- ***ping 60.60.60.2***
- ***ping 10.10.10.1***
- ***ping 10.10.10.2***
- ***ping 10.10.10.3***

- These **ping** commands will **succeed** from **PC1**
- ***ping 20.20.20.1***
- ***ping 192.162.10.2***
- ***ping 192.162.10.1***
- ***ping 50.50.50.2***
- ***ping 50.50.50.3***

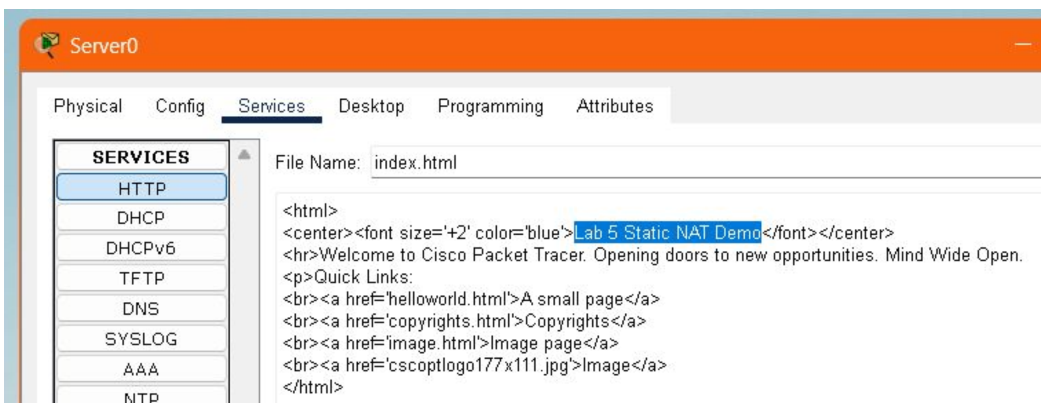
Exp 1: Accessing the Web Service from PC1

Exp 1: Accessing the Web Service on Server0 from PC1

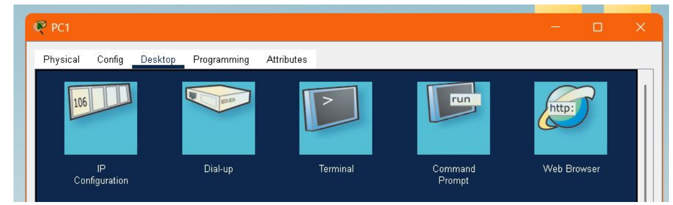
1. Server0 □ Services □ HTTP □ Index.html □ Edit



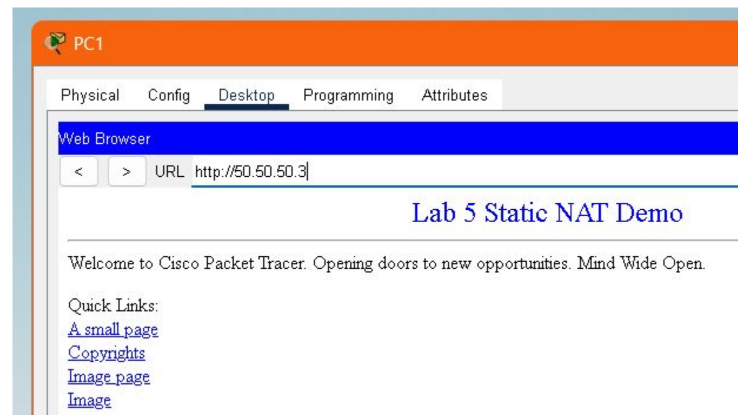
Edit as shown below and Save



1. PC1 □ Desktop □ Web Brower



2. //50.50.50.3 (webserver's external IP)



Lab 2: Summary

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- **Exp 0**: Configuring Static NAT
- **Exp 1**: Accessing the Web Server0 from PC1